

Teaching Exam

Subject – Computer Science

Topic – Computer Network

Lecture No.- 08



By– Rahul Sir

Topics to be Covered

Topic

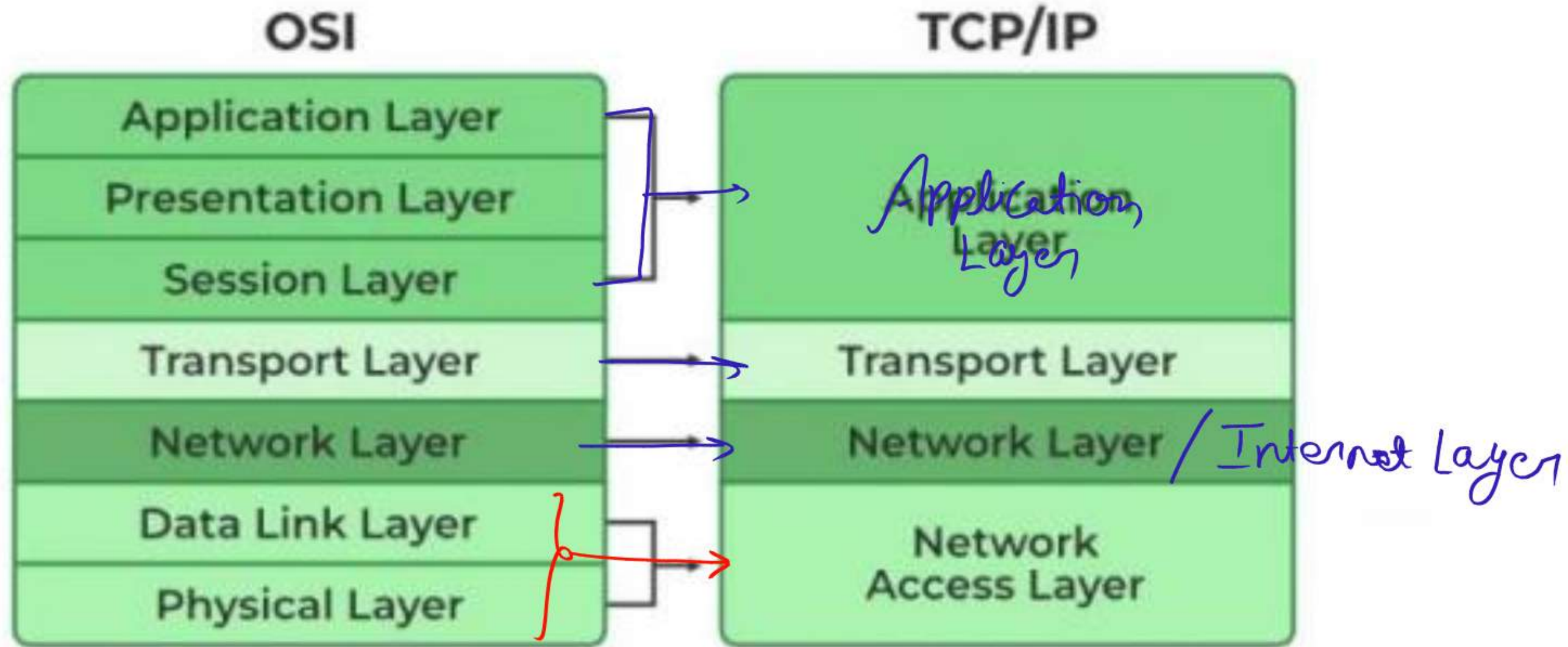
1:-TCP/IP PROTOCOL ✓

2:UDP PROTOCOL ✓

3:-CLIENT-SERVER MODEL ✓

4:- SMTP PROTOCOL AND ITS WORKING MODEL ✓



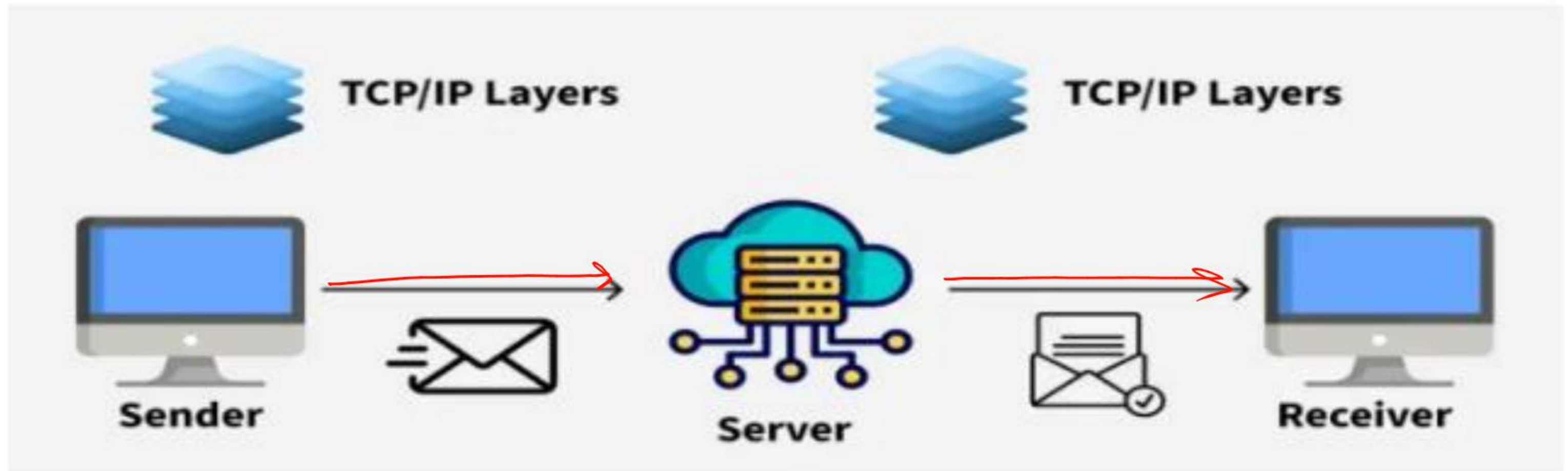


The TCP/IP model is a framework that is used to model the communication in a network. It is mainly a collection of network protocols and organization of these protocols in different layers for modeling the network.

- It has four layers, Application, Transport, Network/Internet and Network Access.
- While the OSI model has seven layers, the 4 layer TCP/IP model is simpler and commonly used in today's Internet and networking systems.

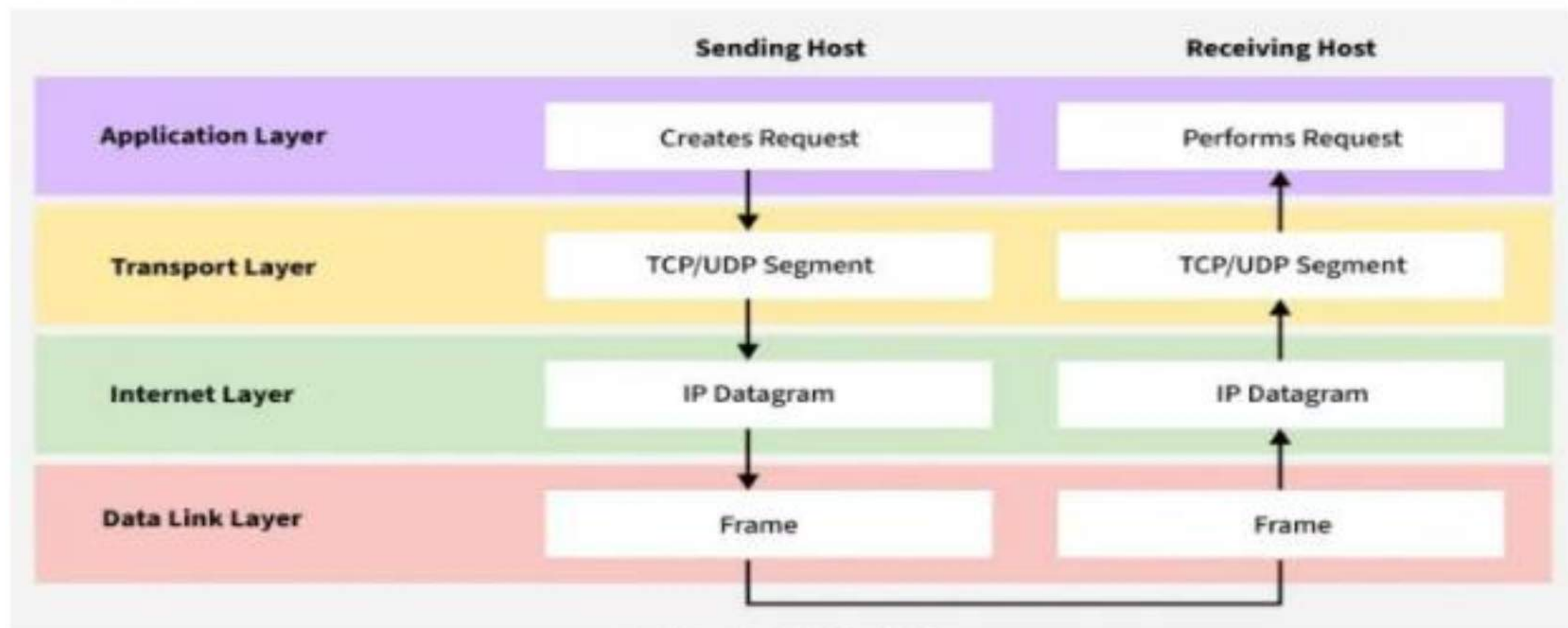
Role of TCP/IP

One of its main goals is to make sure that the data sent by the sender arrives safely and correctly at the receiver's end. To do this, the data is broken down into smaller parts called packets before being sent. These packets travel separately and are reassembled in the correct order when they reach the destination. This helps prevent errors and makes sure the message is complete and accurate.



Working of TCP/IP Model

The working of TCP/IP can be explained with the help of the diagram given below and explained :



When Sending Data (From Sender to Receiver)

- ④ • **Application Layer:** Prepares user data using protocols like HTTP, FTP, or SMTP.
- ③ • **Transport Layer (TCP/UDP):** Breaks data into segments and ensures reliable (TCP) or fast (UDP) delivery.
- ② • **Internet Layer (IP):** Adds IP addresses and decides the best route for each packet.
- ① • **Link Layer (Network Access Layer):** Converts packets into frames and sends them over the physical network.

Segment + Presentation Headers

Segment

Frame + MAC Header + IP Header

Complete frame

frames + MAC + Error Header checking

When Receiving Data (At the Destination)

Data → Binary bit

- ④ • **Link Layer:** Receives bits from the network and rebuilds frames to pass to the next layer.
- ③ • **Internet Layer:** Checks the IP address, removes the IP header, and forwards data to the Transport Layer.
- ② • **Transport Layer:** Reassembles segments, checks for errors, and ensures data is complete.
- ① • **Application Layer:** Delivers the final data to the correct application (e.g., displays a web page in the browser).

4. Network Access Layer

The Network Access Layer is the bottom layer of the TCP/IP model. It deals with the actual physical connection between devices on the same local network like computers connected by cables or communicating through Wi-Fi. This layer makes sure that data can travel over the hardware, such as wires, switches, or wireless signals.

It also handles important tasks like using MAC addresses to identify devices, creating frames (the format used to send data over the physical link), and checking for basic errors during transmission.

3. Internet Layer

The Internet Layer is used for finding the best path for data to travel across different networks so it can reach the right destination. It works like a traffic controller, helping data packets move from one network to another until they reach the correct device. This layer uses the Internet Protocol (IP) to give every device a unique IP address, which helps identify where data should go.

The main job of this layer is routing deciding the best way for data to travel. It also takes care of packet forwarding (moving data from one point to another), fragmentation (breaking large data into smaller parts), and addressing.

2. Transport Layer

The Transport Layer is responsible for making sure that data is sent reliably and in the correct order between devices. It checks that the data you send like a message, file, or video arrives safely and completely. This layer uses two main protocols: TCP and UDP, depending on whether the communication needs to be reliable or faster.

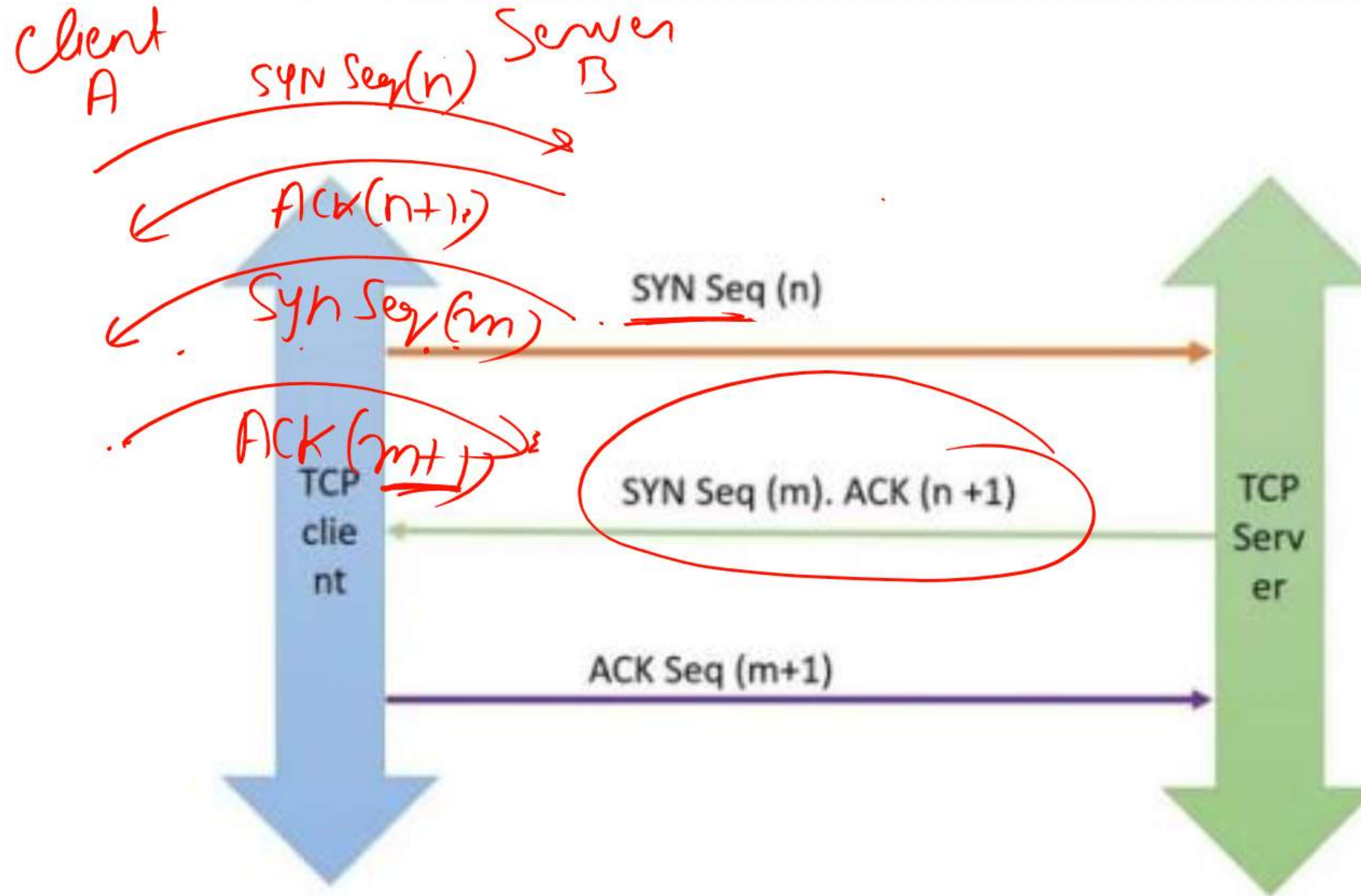
TCP is used when data must be correct and complete, like when loading a web page or downloading a file. It checks for errors, resends missing pieces, and keeps everything in order. On the other hand, UDP (User Datagram Protocol) is faster but doesn't guarantee delivery useful for things like live video or online games where speed matters more than perfect accuracy.

1. Application Layer

The Application Layer is the top layer of the TCP/IP model and the one closest to the user. This is where all the apps you use like web browsers, email clients, or file sharing tools connect to the network. It acts like a bridge between your software (like Chrome, Gmail, or WhatsApp) and the lower layers of the network that actually send and receive data.

It supports different protocols like HTTP (for websites), FTP (for file transfers), SMTP (for emails), and DNS (for finding website addresses). It also manages things like data formatting, so both sender and receiver understand the data, encryption to keep data safe, and session management to keep track of ongoing connections.

Topic- TCP/IP-> Handshaking between TCP-CLIENT/SERVER



Advantages of TCP/IP Model

- **Interoperability** : The TCP/IP model allows different types of computers and networks to communicate with each other, promoting compatibility and cooperation among diverse systems.
- **Scalability** : TCP/IP is highly scalable, making it suitable for both small and large networks, from local area networks (LANs) to wide area networks (WANs) like the internet.
- **Standardization** : It is based on open standards and protocols, ensuring that different devices and software can work together without compatibility issues.
- **Flexibility** : The model supports various routing protocols, data types, and communication methods, making it adaptable to different networking needs.
- **Reliability** : TCP/IP includes error-checking and retransmission features that ensure reliable data transfer, even over long distances and through various network conditions.

Disadvantages of TCP/IP Model

- **Security Concerns** : TCP/IP was not originally designed with security in mind. While there are now many security protocols available (such as SSL/TLS), they have been added on top of the basic TCP/IP model, which can lead to vulnerabilities.
- **Inefficiency for Small Networks** : For very small networks, the overhead and complexity of the TCP/IP model may be unnecessary and inefficient compared to simpler networking protocols.
- **Limited by Address Space** : Although IPv6 addresses this issue, the older IPv4 system has a limited address space, which can lead to issues with address exhaustion in larger networks.
- **Data Overhead** : TCP the transport protocol, includes a significant amount of overhead to ensure reliable transmission.

SSL
TLS

16 byte

8 byte

What is User Datagram Protocol?

User Datagram Protocol (UDP) is one of the core protocols of the Internet Protocol (IP) suite. It is a communication protocol used across the internet for time-sensitive transmissions such as video playback or DNS lookups. Unlike Transmission Control Protocol (TCP), UDP is connectionless and does not guarantee delivery, order, or error checking, making it a lightweight and efficient option for certain types of data transmission.

UDP

Sender



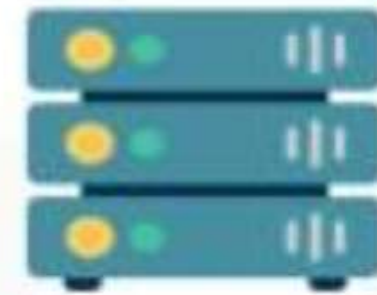
Destination Port	Source Port	Data	Checksum
2	2	2 byte	2 byte

Header
X

lose
check

⇒ 8 byte Packet

Receiver



REQUEST

RESPONSE

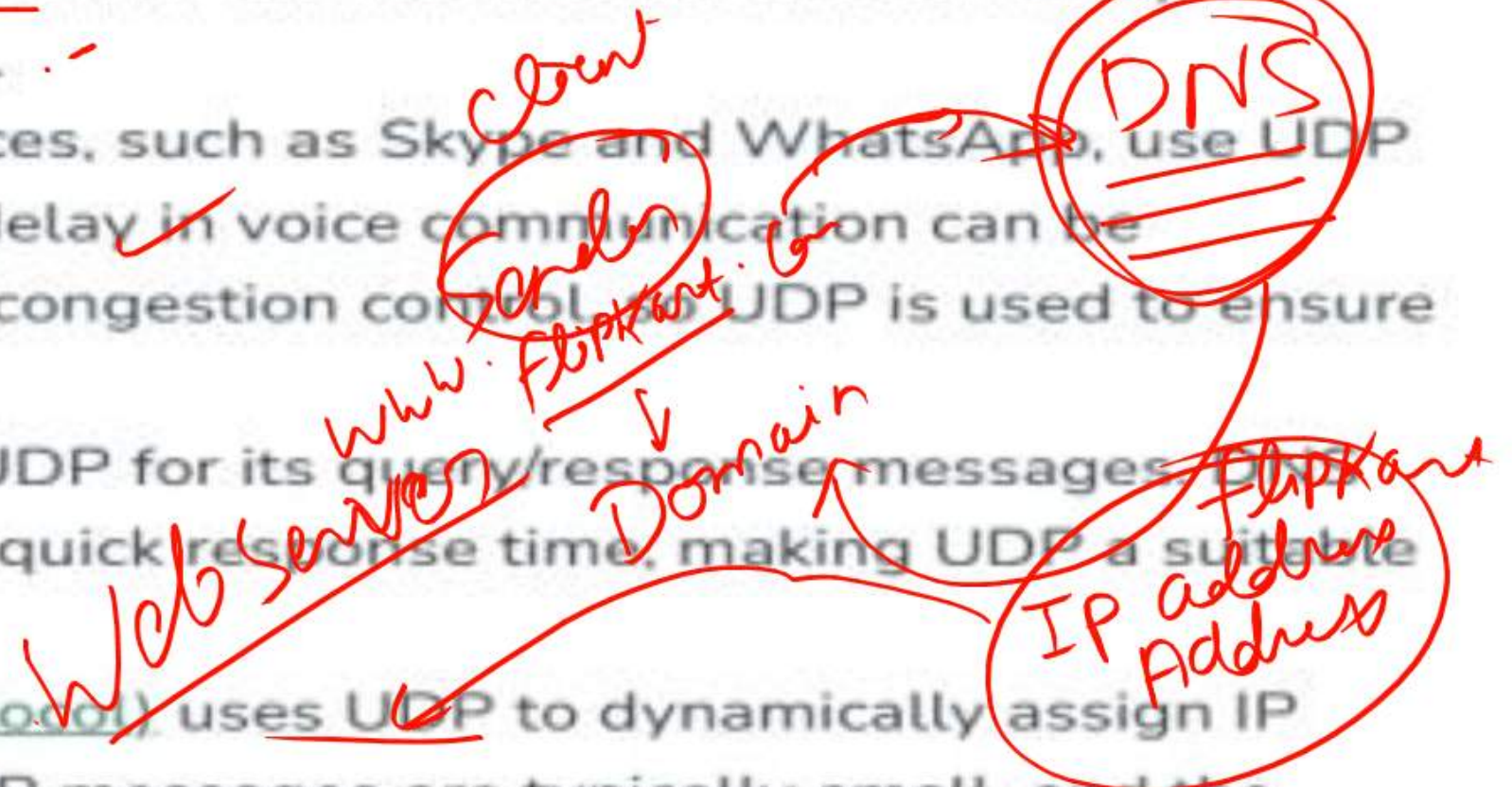
Use Case :- Video Conferencing



- **Source Port**: Source Port is a 2 Byte long field used to identify the port number of the source.
- **Destination Port**: It is a 2 Byte long field, used to identify the port of the destined packet.
- **Length**: Length is the length of UDP including the header and the data. It is a 16-bits field.
- **Checksum**: Checksum is 2 Bytes long field. It is the 16-bit one's complement of the one's complement sum of the UDP header, the pseudo-header of information from the IP header, and the data, padded with zero octets at the end (if necessary) to make a multiple of two octets.

Applications of UDP

- Used for simple request-response communication when the size of data is less and hence there is lesser concern about flow and error control.
- It is a suitable protocol for multicasting as UDP supports packet switching.
- UDP is used for some routing update protocols like RIP (Routing Information Protocol).
- Normally used for real-time applications which can not tolerate uneven delays between sections of a received message.
- VoIP (Voice over Internet Protocol) services, such as Skype and WhatsApp, use UDP for real-time voice communication. The delay in voice communication can be noticeable if packets are delayed due to congestion control. UDP is used to ensure fast and efficient data transmission.
- DNS (Domain Name System) also uses UDP for its query/response messages. DNS queries are typically small and require a quick response time, making UDP a suitable protocol for this application.
- DHCP (Dynamic Host Configuration Protocol) uses UDP to dynamically assign IP addresses to devices on a network. DHCP messages are typically small, and the delay caused by packet loss or retransmission is generally not critical for this application.



Basis	Transmission Control Protocol (TCP)	User Datagram Protocol (UDP)
Type of Service	<u>TCP</u> is a <u>connection-oriented protocol</u> . <u>Connection orientation</u> means that the communicating devices should establish a connection before transmitting data and should close the connection after transmitting the data.	<u>UDP</u> is the <u>Datagram-oriented protocol</u> . This is because there is no overhead for opening a connection, maintaining a connection, or terminating a connection. <u>UDP is efficient for broadcast and multicast types of network transmission.</u>
Reliability	TCP is reliable as it guarantees the delivery of data to the <u>destination router</u> .	The delivery of data to the destination cannot be guaranteed in UDP.

Error checking mechanism	TCP provides extensive error-checking mechanisms. It is because it provides <u>flow control and acknowledgment of data.</u>	UDP has only the basic error-checking mechanism using <u>checksums.</u>
Acknowledgment	An <u>acknowledgment segment</u> is <u>present.</u>	<u>No acknowledgment segment.</u>
Sequence	Sequencing of data is a feature of Transmission Control Protocol (TCP). this means that packets arrive in order at the receiver.	There is <u>no sequencing of data</u> in UDP. If the order is required, it has to be managed by the <u>application layer.</u>
* Speed	TCP is comparatively slower than UDP.	UDP is faster, simpler, and more efficient than TCP.

* Header Length	TCP has a (20-60) bytes variable length header.	UDP has an 8 bytes fixed-length header.
* Weight	TCP is heavy-weight.	UDP is lightweight.
* Handshaking Techniques	Uses handshakes such as SYN, ACK, SYN-ACK	It's a connectionless protocol i.e. No handshake
* Broadcasting	TCP doesn't support Broadcasting.	UDP supports Broadcasting.
Protocols	TCP is used by <u>HTTP</u> , <u>HTTPS</u> , <u>FTP</u> , <u>SMTP</u> and <u>Telnet</u> .	UDP is used by <u>DNS</u> , <u>DHCP</u> , <u>TFTP</u> , <u>SNMP</u> , <u>RIP</u> , and <u>VoIP</u> .

Advantages of UDP

- **Speed:** UDP is faster than TCP because it does not have the overhead of establishing a connection and ensuring reliable data delivery.
- **Lower latency:** Since there is no connection establishment, there is lower latency and faster response time.
- **Simplicity:** UDP has a simpler protocol design than TCP, making it easier to implement and manage.
- **Broadcast support:** UDP supports broadcasting to multiple recipients, making it useful for applications such as video streaming and online gaming.
- **Smaller packet size:** UDP uses smaller packet sizes than TCP, which can reduce network congestion and improve overall network performance.
- User Datagram Protocol (UDP) is more efficient in terms of both latency and bandwidth.

Disadvantages of UDP

- **No reliability:** UDP does not guarantee delivery of packets or order of delivery, which can lead to missing or duplicate data.
- **No congestion control:** UDP does not have congestion control, which means that it can send packets at a rate that can cause network congestion.
- **Vulnerable to attacks:** UDP is vulnerable to denial-of-service attacks, where an attacker can flood a network with UDP packets, overwhelming the network and causing it to crash.
- **Limited use cases:** UDP is not suitable for applications that require reliable data delivery, such as email or file transfers, and is better suited for applications that can tolerate some data loss, such as video streaming or online gaming.

Client

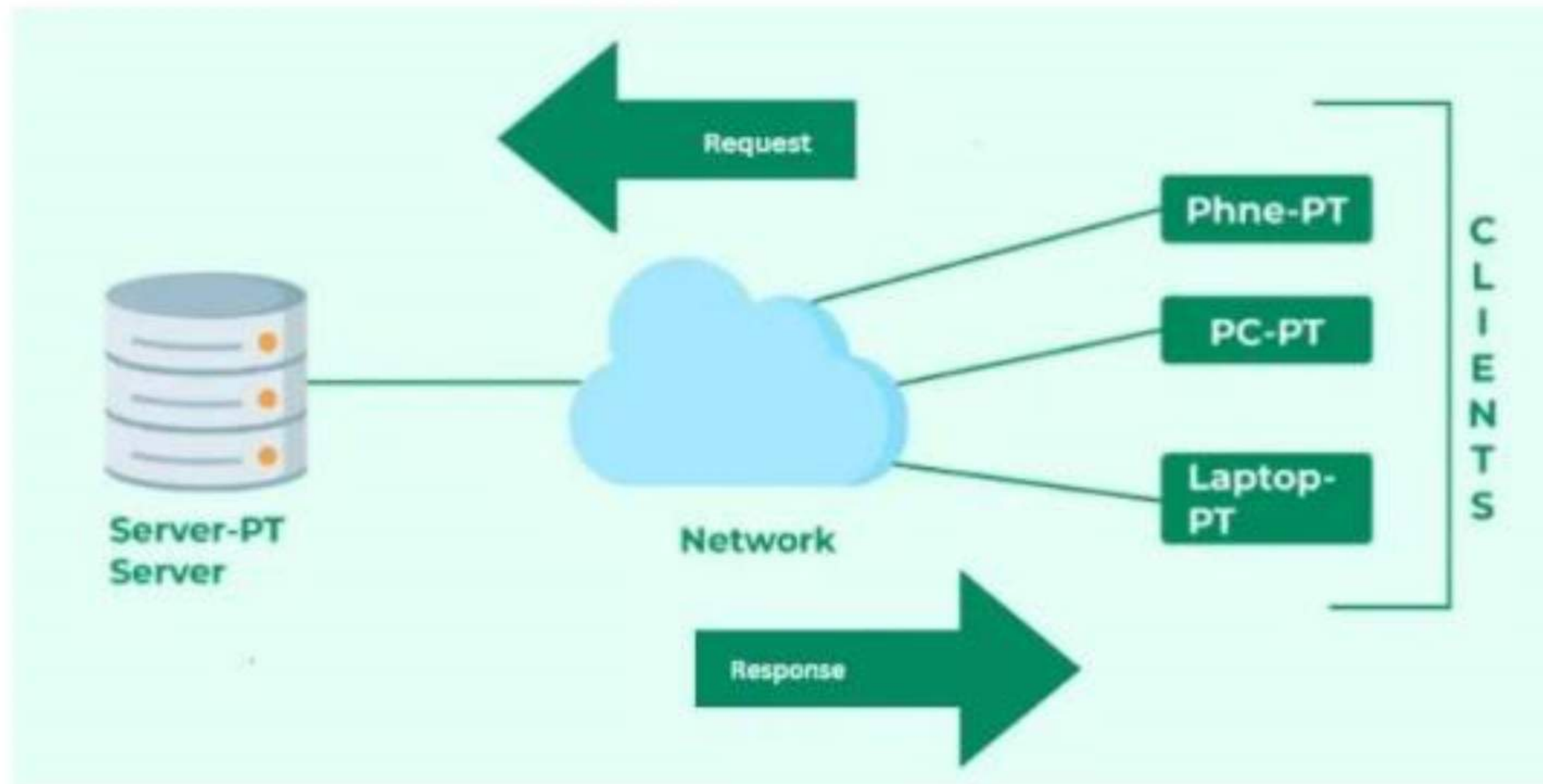
Client is machine which is Seeking (demand) Service from Server

When we talk about a "Client," it refers to a device that requests and receives services from a server. The client is the entity that initiates communication, asking for data or resources from the server. For example, web browsers like Google Chrome, Mozilla Firefox or Safari are common client applications that request data from a server to render web pages.

Server

Server is machine which Seeking (demand) request from different Client

A Server, on the other hand, is a remote computer or system that provides data, resources or services to clients. It listens to incoming client requests, processes them and sends the required information back. A server can handle multiple client requests simultaneously.





How the Browser Interacts With the Servers?

The process of interacting with servers through a browser involves several steps. Here's a breakdown of the steps taken when you enter a URL in a browser and receive the website data:

- 1. User Enters the URL (Uniform Resource Locator):** The user types a website address (e.g., `www.example.com`) into the browser's address bar.
- 2. DNS (Domain Name System) Lookup:** The browser sends a request to the DNS server to resolve the human-readable URL into an IP address (since computers use IP addresses to identify and connect to each other).
- 3. DNS Server Resolves the Address:** The DNS server looks up the domain name and returns the IP address of the web server hosting the requested website.
- 4. Browser Sends HTTP/HTTPS Request:** The browser sends an HTTP/HTTPS request to the IP address of the web server to fetch the website's data. HTTP (HyperText Transfer Protocol) or HTTPS (the secure version) is the protocol used for communication between the browser (client) and the web server (server).

- 5. Server Sends Website Files:** The server processes the request and sends the necessary website files (HTML, CSS, JavaScript, images, etc.) back to the browser.
- 6. Rendering the Website:** The browser renders the files and displays the website to the user. This rendering process involves several components working together:
- **DOM (Document Object Model) Interpreter:** Processes the HTML structure and build tree-like representation of the page.
 - **CSS Interpreter:** Applies styles to the HTML elements.
 - **JS Engine:** Executes JavaScript code for interactivity. Most modern JavaScript engines use Just-In-Time (JIT) compilation to optimize performance.

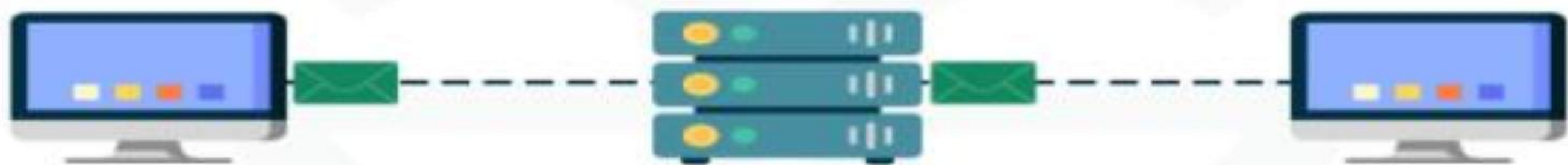


SMTP represents **Simple Mail Transfer Protocol**. SMTP is a set of interaction guidelines that allow the software to transmit electronic mail over the internet, referred to as Simple Mail Transfer Protocol.

The main objective of SMTP is used to set up communication rules between servers. The servers have a way of recognizing themselves and announcing what kind of communication they are trying to perform. They also have a way of handling errors such as incorrect email address. For example, if the recipient address is wrong, then receiving a server reply with an error message of some kind.

To send an email, the client opens a TCP connection to the SMTP server. The server, which is always listening on port 25, initiates the connection as soon as it detects a client. Once the TCP connection is established, the client sends the email across the connection.

SMTP

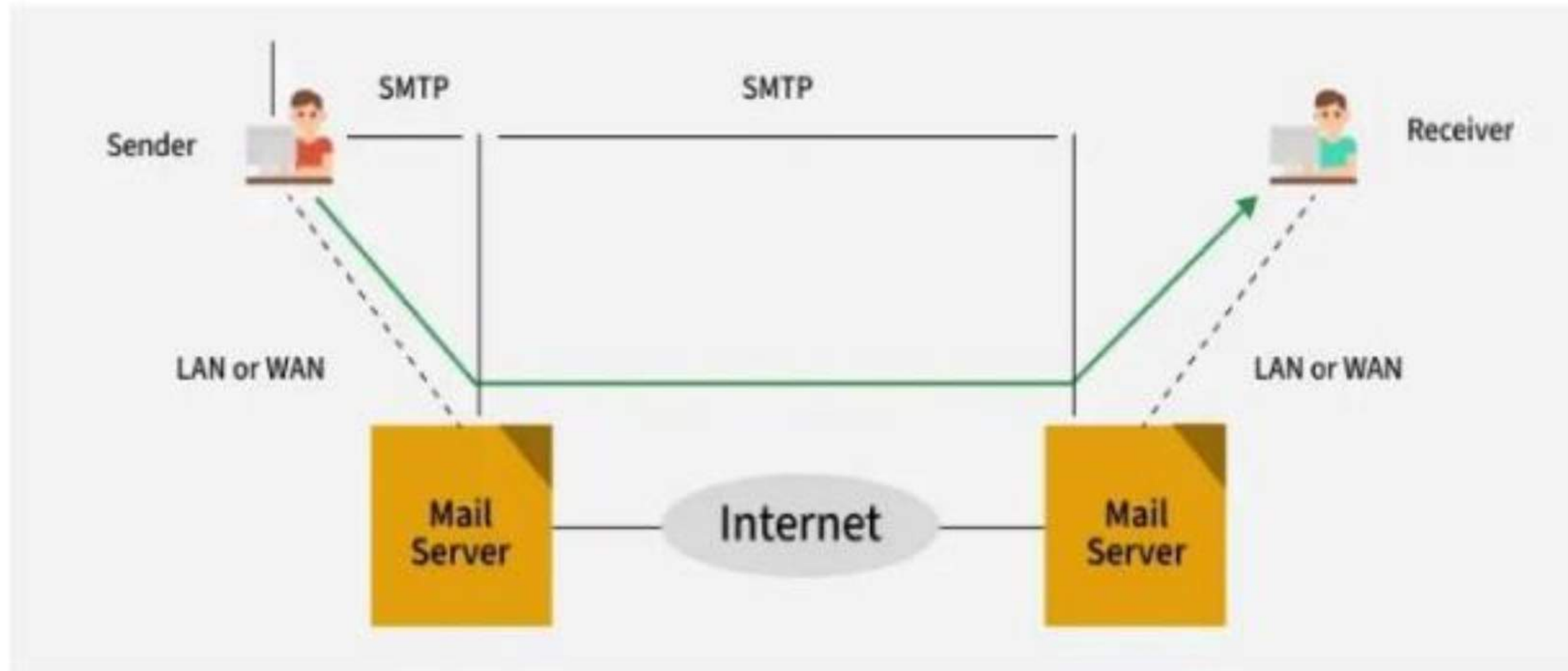


Types of SMTP Protocol

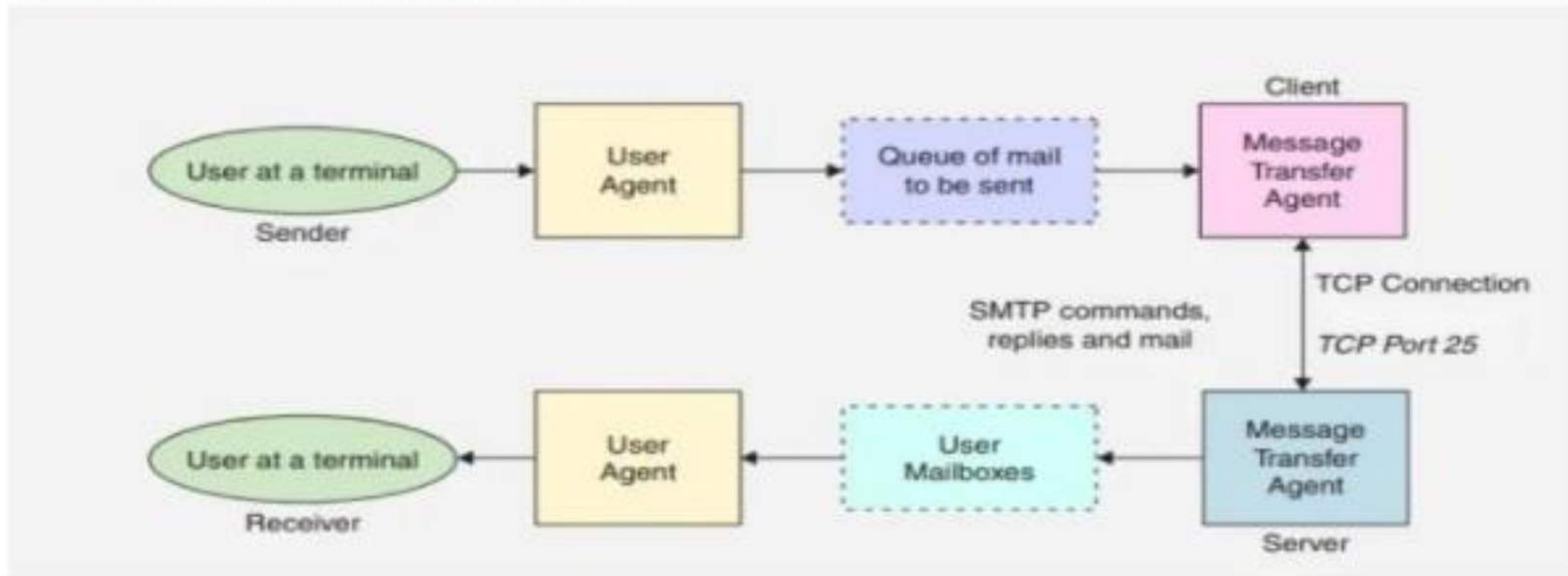
The SMTP model supports two types of email delivery methods: **end-to-end** and **store-and-forward**.

- **End-to-end** delivery is used between organizations. In this method, the email is sent directly from the sender's SMTP client to the recipient's SMTP server without passing through intermediate servers.
- **Store-and-forward** is used within organizations that have TCP/IP and SMTP-based networks. In this method, the email may pass through several intermediate servers (Message Transfer Agents, or MTAs) before reaching the recipient.

With **end-to-end** delivery, the SMTP client waits until the email is successfully copied to the recipient's SMTP server before sending it. This is different from the **store-and-forward** method, where the email might stop at multiple intermediate servers before reaching its destination. In store-and-forward systems, the sender is notified as soon as the email reaches the first server, not the final destination.



Model of SMTP System



THANK YOU